

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

Student Startup Crowdfunding Platform

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Selected Architecture: Layered Architecture

Architecture Selection

We chose **Layered Architecture** for the Student Startup Crowdfunding Platform because the system has clearly separated responsibilities such as user interaction, campaign management, payment and pledge processing, milestone verification, fund release, and data storage.

Reason 1 — Separation of Responsibilities

Layered Architecture provides clear separation between the presentation, business logic, payment processing, milestone management, and data storage responsibilities. This makes the crowdfunding platform easier to understand, maintain, test, and modify.

Reason 2 — Suitability for the System

The platform does not require the operational complexity of a Microservices Architecture. A layered structure is suitable for a university crowdfunding platform because the major system functions can be organized into well-defined components while maintaining a relatively simple deployment and management structure.

Security Advantage

Sensitive operations such as payment processing and milestone-based fund release can be handled within dedicated components and protected by validation and access-control mechanisms. Payment information can be handled through secure or tokenized payment processing, reducing unnecessary exposure of sensitive payment data.

Performance Benefit

Layered Architecture avoids the additional network and operational overhead associated with multiple independently deployed microservices. Direct interactions between the system components can support responsive campaign, pledge, milestone, and fund-release operations while meeting the system's performance needs.